

Vyankatesh Kulkarni

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ABOUT ME

Software Engineering student with hands-on experience contributing to backend systems and AI-powered applications. Worked on production software involving APIs, databases, automation, and AI systems, with an interest in backend engineering, distributed systems, and applied AI.

EDUCATION

BE, Information Technology (IT)

Pune Institute of Computer Technology (PICT) (GPA: 9.33/10)

• **Achievements:** 9.80+ SGPA in 5 of 6 semesters

Sep 2023 - Jul 2027

Pune

Work Experience

Databahn.ai | Software Engineering Intern

Feb 2026 - Aug 2026

- Developed features for a **Go-based AI workflow platform**, implementing component definitions and core backend logic.
- Engineered a Spring Boot API **abstraction layer for 220+ source and destination configurations**, with dynamic Swagger schemas, validation, and public-to-private field transformation.
- Developed Pytest-based API and end-to-end automation, increasing backend automation coverage from **8% to 47% in 3 weeks by automating 270+ TestRail cases**.
- Optimized API automation polling by increasing the polling interval from **10s to 45s**, reducing API requests by **~3X**.
- Optimized end-to-end test execution by chaining test cases and reusing shared infrastructure across scenarios, eliminating redundant resource creation and activation waits and saving **~5-6 minutes per test case**.
- Built an **agent workflow to classify miscategorized TestRail cases** using test descriptions, expected results, and reproduction steps, enabling identification and automation of backend/API test cases.

Bharti Hospitals (Dental) | Freelancer

Jan 2026 - May 2026

- **Architected and delivered** a live AI-powered dental diagnosis platform processing **8 intraoral images and 14 clinical questions** to generate tooth-wise disease mapping and automated clinical reports.
- Built a **multistage AI pipeline** combining **MobileNet-based view validation**, **YOLO-based tooth detection**, and **8 specialized VLM agents** for tooth-level disease classification.
- Designed image and case-data handling using **Amazon S3 and Redis**, storing image references and questionnaire state by case ID to minimize large image payloads in backend memory.
- Implemented **real-time diagnosis progress streaming using SSE** across detection, diagnosis, and report-generation stages.
- Achieved **~98% view-validation accuracy**, **~95% tooth-detection accuracy**, and **~85% disease-classification accuracy**, with automated **DMFT scoring and odontogram-based clinical reports**.
- Built **medical camp management functionality**, enabling administrators to create camps, onboard patients through camp credentials, and organize and filter generated reports by disease severity.

PROJECTS

Chainvote - Blockchain based voting platform

- Built the **backend for a blockchain-based voting platform**, supporting community creation, member onboarding, election management, and voter eligibility.
- Implemented **configurable community schemas**, allowing administrators to define custom fields and validation rules for member registration.
- Developed **dynamic eligibility and filtering logic** to select candidates and voters based on administrator-defined community attributes and election requirements.

NyayaGPT - Agentic AI-powered Legal chatbot

- Built a **two-agent legal reasoning pipeline** to analyze user queries, determine retrieval requirements, and generate context-aware responses.
- Implemented **Case Law Mode** using embeddings and cosine-similarity search over **10K+ legal case records**, retrieving the **top-5 relevant cases** for a query.
- Integrated **external case-law API retrieval** into the agent workflow, combining fetched case information with locally retrieved context for response generation.

SKILLS

- **Languages:** C++, Java, JavaScript, Python
- **Computer Science Fundamentals:** DSA, OS, OOP, DBMS, System Design
- **Backend & Frameworks:** Node.js, REST APIs, Spring Boot, FastAPI, Pytest, Go
- **Databases & Storage:** MongoDB, PostgreSQL, MySQL, Redis, Amazon S3
- **AI/ML:** Deep Learning, Computer Vision, Machine Learning, Agentic AI
- **Security and API tools:** JWT, Swagger
- **Tools & Infrastructure:** Docker, Jenkins, GitHub, Postman, TestRail, Jira, Confluence

Achievements

- **550+ DSA Problems:** Solved 550+ algorithmic problems across LeetCode, CodeChef and CSES
- **LeetCode (1671):** Achieved a maximum rating of 1671

CERTIFICATIONS & LEARNINGS

- Deep Learning Specialization - DeepLearning.AI
- Software Design & Architecture (In Progress) - University of Alberta
- Full Stack Web Development - 100xDevs
- Machine Learning & Data Science Masterclass - Udemy

Extracurricular

Chess | *State-level competitor (Top 10)*